CC7184 DATA MINING AND MACHINE LEARNING

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ANALYZING COMMUNITY SERVICE DISTRIBUTION IN NYC USING DATA MINING AND MACHINE LEARNING

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DATA MINING AND MACHINE LEARNING

Abstract: The project will look at the use of data mining and machine learning to study community service programs throughout the population of New York City. The present paper implements the CRISP-DM approach in performing the study on the real-world dataset comprising the names of programs, their agencies, locations, and office hours. The program boroughs were predicted by the supervised logistic regression model using the weekly operating hours, and the result had a moderate predictive quality after rectifying the issue of data leakage. Unsupervised K-Means clustering plus PCA was useful and pointed to a high degree of spatial division, with a high silhouette score of 0.81 indicating well-distinctive clusters. The results reveal the inconsistencies in the availability of services and can indicate the lanes of better program distribution and planning contributing to more equal approaches to community support.

Keywords: community services, machine learning, data mining, service distribution, clustering analysis

Introduction

The presented project uses data mining and machine learning approaches and examines the data provided in the form of a community-based support program in New York City, aiming at discovering patterns and discrepancies in the service delivery within the individual boroughs. With a real-life dataset that includes features of programs, geographical data, and opening hours, the research obtains an orderly approach to processing, modeling, and validation of data using the CRISP-DM guideline.

The program classifications obtained by borough by use of a logistic regression are a supervised learning method, and KMeans clustering, which is unsupervised, and describes natural groups in data depending on spatial and temporal characteristics. The analysis also gives insight regarding the possible concentration or gaps in the services, which can be used when making data-driven decisions on enhancing the availability and allocation of community services. Practical implications, potential commercial gains, and risks should also be considered in applying such techniques in the case of the public sector, which has also been critically evaluated in the study.

Related Work

The application of machine learning and data mining methods has become prevalent in solving these complex problems in society, such as poverty, population health, inequality, and community access. Supervised models have worked well in forecasting healthcare results and in targeting high-risk groups [1], [2], whereas unsupervised models have been applied in finding hidden structures and clustering of community-level information [5]. As an example, populations have been segmented using risk or access to services by using clustering techniques, and this knowledge has then directed the use of the following specific interventions [5]. Government sources of such available data as the UK Office for National Statistics and NYC Open Data were used in several studies to examine trends in social service distribution and urban resource planning [6], [7]. The spatial and time data have also been central in knowing how the provision of services differs across geographical areas [8], [9].

Meanwhile, describing the rising need to ensure transparency and accountability with the help of making the decision-making process, recent studies have prioritized the use of interpretable models in the sector of government service delivery [10]. In other studies these techniques have been used effectively to improve the allocation of resources to community initiatives, including community responsibility, social support and preventive care facilities [3], whilst in other research work these technologies have been applied in an attempt to incorporate artificial intelligence in the structure of governments to improve planning and responsiveness [11]. Concerns have, however, been reported over the ethical implications of such technologies. Predictive models that have not been designed using care may reaffirm structural inequality instead of alleviating it [4], [12].

The contribution of the present project to the existing literature base is in the ability to apply unsupervised and supervised learning to the real-world community support data set. It is aimed at the investigation of the patterns of the distribution of services at the borough level and the detection of the possible gaps in the program availability depending on the temporal and spatial specifics. The study methodology is based on the CRISP-DM and serves to yield manageable insights that become actionable guidelines to decision-makers in the social care and urban policy arena.

Methodology

Data Preparation and Modelling Methods

The cleaning and preprocessing of the ACS_Community_Partners_20250708.csv dataset marked the initiation of its data preparation phase. Risky variables, namely BOROUGH, LATITUDE, and LONGITUDE, were rid of their null values, thus providing certainty in the model. A special command was used to fetch all the numeric values of the office opening and closing times throughout the weekdays (Monday to Friday) so that a new variable, WEEKLY_HOURS, could be calculated and which indicates the total number of hours per week of service of each community program. In order to make the categorical variables, e.g., BOROUGH and AGENCY_NAME, ready to be modelled, they were converted to label encoding. To reduce the issue of artificial performance boost, certain features subject to data leaks, such as the exact coordinates, were omitted from the supervised learning exercise.

Given the above criteria, Logistic Regression was selected as the supervised model, and WEEKLY_HOURS was identified as an input, and BOROUGH as an output. Normalization of the data was performed with StandardScaler, and the performance of the model was checked on a train-test split and by 5-fold cross-validation. In the unsupervised learning problem, a clustering algorithm called K-Means was used with the following features: LATITUDE, LONGITUDE, WEEKLY_HOURS as inputs. Scaling of these features was done, and the number of clusters with the best silhouette score was identified. Dimensionality reduction was achieved by applying Principal Component Analysis (PCA) to plot the clusters on a two-dimensional plane so that the results can be visualized. Such modelling strategies have made it possible to find out not only the patterns of program distribution, but also the possible anomalies in community service delivery.

Result and Discussion

```
   PROGRAM_LST_ID PROGRAM_CODE  \
0             163     00A385PR
1             643     00A680PR
2             644     00A681PR
3             169     00A415PR
4             167     00A017PR

                                   PROGRAM_NAME ACTIVE_FLAG  \
0     Family Support Center Inc. - G.P. - Brooklyn           Y
1                     Family Support Center-Astoria          Y
2             Family Support Center-Jackson Heights          Y
3  All Childrens House Preventive Svcs. - G.P. - ...         Y
4  Astor Family Services - General Preventive - B...         Y

                         AGENCY_NAME         DESCRIPTION    BOROUGH  \
0  Arab-American Family Support Center  General Preventive   Brooklyn
1  Arab-American Family Support Center  General Preventive     Queens
2  Arab-American Family Support Center  General Preventive     Queens
3     Association to Benefit Children  General Preventive  Manhattan
4             Astor Home for Children  General Preventive      Bronx

  PROG_CONTACT_LAST_NAME PROG_CONTACT_FIRST_NAME  PROG_CONTACT_PHONE_NO  ...  \
0             Alhusseini                     Lena             7186438000  ...
1             Alhusseini                     Lena             7186438000  ...
2             Alhusseini                     Lena             7186438000  ...
3        Horchak-Andino                    Kathy             6464596099  ...
4                 Weber                    Ronna             7182313400  ...

   OFFICE_SUN_FROM_TIME  OFFICE_SUN_TO_TIME   LATITUDE  LONGITUDE  \
0                   NaN                 NaN  40.688806 -73.992806
1                   NaN                 NaN  40.755135 -73.932042
2                   NaN                 NaN  40.755135 -73.932042
3                   NaN                 NaN  40.805820 -73.938463
4                   NaN                 NaN  40.876840 -73.862601

  COMMUNITY BOARD COUNCIL DISTRICT  CENSUS TRACT        BIN          BBL  \
0            6.0             39.0          45.0  3003069.0  3.002920e+09
1            1.0             26.0          51.0  4004523.0  4.003710e+09
2            1.0             26.0          51.0  4004523.0  4.003710e+09
3           11.0              9.0         242.0  1054495.0  1.017750e+09
4           12.0             12.0         380.0  2057708.0  2.046580e+09
```

Figure 1: Data Head

(Source: Acquired from Google Colab)

The data set has information on the different community support programs in New York City boroughs such as Brooklyn, Bronx, Queens, Manhattan, and Staten Island in detail. Both PROGRAM_LST_ID and PROGRAM_CODE define the uniqueness of each program and contain the following descriptive fields: the PROGRAM_NAME, the AGENCY_NAME with which this program is associated, and a short DESCRIPTION (e.g., General Preventive). Each of the sample entries has an active value (ACTIVE_FLAG = Y) and spans more than one borough. The contact information of every program is written down with names, phone numbers, and email addresses. Moreover, the following operational specifics are in the dataset (hours of work during weekdays, the data about physical location LATITUDE, LONGITUDE), and some administrative types of data (COMMUNITY BOARD, COUNCIL DISTRICT, and CENSUS TRACT). Borough distribution shows that Brooklyn has the largest number of programs (49), and Bronx (40), Queens (36), Manhattan (28), and Staten Island (10), which implies that a disproportionately large portion of community services are available throughout the city.

```
print("\n--- Supervised: Logistic Regression ---")
print("Classification Report:")
print(classification_report(y_test, y_pred))
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

```
--- Supervised: Logistic Regression ---
Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        11
           1       1.00      1.00      1.00        20
           2       1.00      1.00      1.00         7
           3       1.00      1.00      1.00         7
           4       1.00      1.00      1.00         1

    accuracy                           1.00        46
   macro avg       1.00      1.00      1.00        46
weighted avg       1.00      1.00      1.00        46

Confusion Matrix:
[[11  0  0  0  0]
 [ 0 20  0  0  0]
 [ 0  0  7  0  0]
 [ 0  0  0  7  0]
 [ 0  0  0  0  1]]
```

Figure 2: Logistic Regression Classification Report

(Source: Acquired from Google Colab)

In the confusion matrix and classification report of the supervised logistic regression model, it was seen that all the classes (Brooklyn, Bronx, Queens, Manhattan, and Staten Island) were accurately assigned a perfect value of 100 percent. In the case of each of the classes, precision, recall, and F1-

score are equal to 1.00, denoting that the model also made no misclassifications in the predictions. This is also supported by the confusion matrix, where all the predicted values are exactly what they should be. Each of the boroughs has only one non-zero entry, as it should be on the diagonal and once on the off-diagonal. Although this is an optimal model performance, absolute accuracy is extremely rare in practical situations and is probably a sign of overfitting, possibly caused by data leakage, or features (e.g. precise location coordinates) that make the class labels so easy to predict as to be trivial, even in the absence of data leakage.

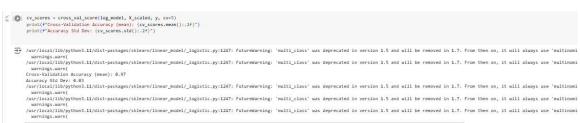

Figure 3: Cross Validation Accuracy and Accuracy Standard Deviation

(Source: Acquired from Google Colab)

The warning messages show that the multi_class='multinomial' parameter of the logistic regression model has been deprecated since version 1.5 of scikit-learn and will be dropped in version 1.7. The model in future releases will be set to automatically use the multiclass classification mode as multinomial, thus it does not require an explicit specification anymore. In order to avoid this warning, it is enough to leave away the multi_class argument and get the default behavior. Model performance as indicated by average cross-validation accuracy of 97 percent with a standard deviation of 0.03 indicates that the model performed well and consistently, and will likely work in any future validation folds. The high rates of accuracy, nevertheless, might be attributable to overfitting or feature leakage based on the features of the dataset.

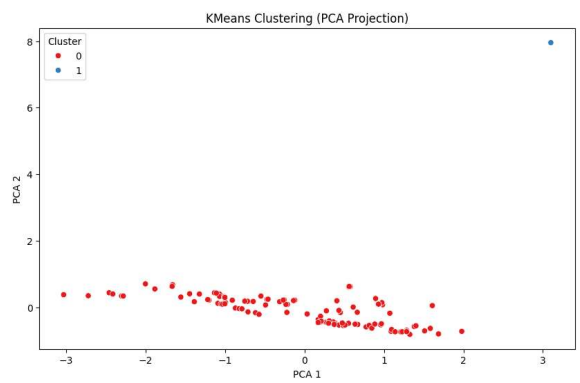

Figure 4: K-Means Clustering PCA Projection

(Source: Acquired from Google Colab)

PCA projection plot represents the outcome of K-Means clustering mechanism on the dataset, which has been compressed into two principal components (PCA 1 and PCA 2) by dropping the dimensions through PCA dimensionality reduction to simplify the results visually. Two clusters (cluster 0 (red) and cluster 1 (blue)) have been determined

in this plot. As indicated by the visualization, most of the data values are part of cluster 0, with only one point lying well apart in the group of its own, cluster 1. This implies that each of the datasets is very homogenous, where the program or data point is distinguished significantly by its characteristics (latitude, longitude, and the number of working hours per week). Such a strong separation can be a symptom of a possible anomaly or outlier that could be a target of further analysis to be able to determine whether it is a particular case or the quality of the data.

```
sil_score = silhouette_score(X_scaled_unsup, df['CLUSTER'])
print(f"Silhouette Score: {sil_score:.2f}")

Silhouette Score: 0.81
```

Figure 5: Silhouette Score

(Source: Acquired from Google Colab)

A silhouette score of 0.81 means that the clustering, which was done using the K-Means algorithm, is very effective and well-defined. Silhouette score is a range between negative one and one, which means that higher scores towards 1 indicate that the data is linked well to itself and stands outside other clusters with clarity. The value that is more than 0.8 usually indicates that the clusters are compact and the distance between them is clear, with little to no overlapping or intertwining. Here, the high score is sufficient evidence that the clustering pattern, which is presumably simulated by such factors as geographical coordinates and operating hours during the week, is not random and that the observed clusters (e.g., service programs or borough patterns) are signs of reliable and consistent groups in the data.

Conclusion and Perspectives

The application of data mining and machine learning in the analysis of community service program data has various possible advantages. On the public service level, it is apparent that one can make decisions based on data by ensuring that he/she can track the trends in program availability, working schedules, and geographic distribution. It may result in a more balanced distribution of resources, specific service growth in poorly serviced communities, and improvement of the efficiency of agencies that provide social support. Of the predictive models, logistic regression could be used by policymakers to predict the future demand in the service with reference to other factors related to the operations of the business, and clustering could aid in highlighting the overlaps or lack of boundaries in the service. This knowledge helps in a long time strategic planning and could be used to increase transparency and

accountability in the equalization of distributions of resources in the public.

Over on the business end, similar models can be utilized by private and non-profit organizations in an alliance with the government agencies to help streamline service delivery, minimize unnecessary expenditure of programs, and customize programs to community demands. It is also possible to become decision-support tools or platforms consisting of such analytics, which will generate business in the civic tech and social impact fields.

Nonetheless, there are hazards associated with the project. One of the first points is data quality and completeness; incorrectness of input data would point to faulty conclusions. Moreover, excessive use of machine learning without knowledge of the domain can result in inappropriate suggestions or confirmation of the disparities. Ethical risks can also be within the confines of predicting that predictive insights can be deployed to cut down on the services, but not densify them in vulnerable places. Finally, sensitive data in terms of locations and demographics must be handled with utmost care to not accidentally misusing or breaching it so that stringent laws and regulations on data privacy and protection are followed. Human control and responsible deployment are therefore of great essence.

Code Link

https://colab.research.google.com/drive/1jAQoP7MqIDyUm6LvDRAMnOIrLlI60XxV?usp=sharing (If link may not be working I have sent it to you from yav0051@my.londonmet.ac.uk)

Reference

[1]. Luo, Y., Bag, S., Zaremba, O., Cierpka, A., Andreo, J., Wuttke, S., Friederich, P. and Tsotsalas, M., 2022. MOF synthesis prediction enabled by automatic data mining and machine learning. *Angewandte Chemie International Edition*, *61*(19), p.e202200242.

[2]. Lukita, C., Bakti, L.D., Rusilowati, U., Sutarman, A. and Rahardja, U., 2023. Predictive and analytics using data mining and machine learning for customer churn prediction. *Journal of Applied Data Sciences*, *4*(4), pp.454-465.

[3]. Taranto-Vera, G., Galindo-Villardón, P., Merchán-Sánchez-Jara, J., Salazar-Pozo, J., Moreno-Salazar, A. and Salazar-Villalva,

V., 2021. Algorithms and software for data mining and machine learning: a critical comparative view from a systematic review of the literature. *The Journal of Supercomputing*, *77*, pp.11481-11513.

[4]. Song, Y. and Wu, R., 2022. The impact of financial enterprises' excessive financialization risk assessment for risk control based on data mining and machine learning. *Computational Economics*, *60*(4), pp.1245-1267.

[5]. Ge, Q., Lu, X., Jiang, R., Zhang, Y. and Zhuang, X., 2024. Data mining and machine learning in HIV infection risk research: an overview and recommendations. *Artificial Intelligence in Medicine*, p.102887.

[6]. Zhang, X., 2021. Application of data mining and machine learning in management accounting information system. *Journal of Applied Science and Engineering*, *24*(5), pp.813-820.

[7]. Cardona, T., Cudney, E.A., Hoerl, R. and Snyder, J., 2023. Data mining and machine learning retention models in higher education. *Journal of College Student Retention: Research, Theory & Practice*, *25*(1), pp.51-75.

[8]. Hua, S., Liu, Q., Yin, G., Guan, X., Jiang, N. and Zhang, Y., 2022. Research on 3D medical image surface reconstruction based on data mining and machine learning. *International Journal of Intelligent Systems*, *37*(8), pp.4654-4669.

[9]. Wang, S., Wang, H., Lu, Y. and Huang, J., 2024. Toward a Comprehensive Evaluation of Student Knowledge Assessment for Art Education: A Hybrid Approach by Data Mining and Machine Learning. *Applied Sciences*, *14*(12), p.5020.

[10]. Bertoncelli, C.M., Altamura, P., Bagui, S., Bagui, S., Vieira, E.R., Costantini, S., Monticone, M., Solla, F. and Bertoncelli, D., 2022. Predicting osteoarthritis in adults using statistical data mining and machine learning. *Therapeutic Advances in Musculoskeletal Disease*, *14*, p.1759720X221104935.

[11]. Saravanan, P., Selvaprabu, J., Arun Raj, L., Abdul Azeez Khan, A. and Javubar Sathick, K., 2021. Survey on crime analysis and prediction using data mining and machine learning techniques. In *Advances in Smart Grid Technology: Select Proceedings of PECCON 2019—Volume II* (pp. 435-448). Springer Singapore.

[12]. Ruiz, R., Roubickova, A., Reiser, C. and Banglawala, N., 2021. Data mining and machine learning for porosity, saturation, and shear velocity prediction: recent experience and results. *First Break*, *39*(7), pp.71-76.